

Introduction

- Enterprise policies are frequently updated, creating changes that can affect related policies, rules, and procedures.
- The proposed system automatically detects changes between policy versions and identifies potentially affected clauses and policies.
- A PolicyGraph is used to trace relationships and propagate the impact of a policy change across connected documents.
- RAG retrieves supporting policy evidence, while an LLM generates an evidence-grounded explanation of the identified impacts.

Objectives

- Detect and classify meaningful changes between different versions of enterprise policy documents.
- Identify affected policies and clauses by analyzing explicit and semantic relationships between policy content.
- Rank potential impacts using relevance and confidence scores supported by graph-based propagation and evidence retrieval.
- Generate explainable, evidence-grounded impact analysis to support compliance, governance, and policy-management decisions.

Scope

- Covers enterprise policy documents, their versions, clauses, cross-references, and related procedural content.
- Supports change types such as added, removed, and modified clauses and traces their potential downstream impact.
- Combines PolicyGraph-based relationship analysis with RAG-based evidence retrieval for impact assessment.
- Provides ranked impact results, supporting evidence, confidence scores, and LLM-generated explanations through a web interface.

Literature Review

- Traditional policy and document analysis approaches rely heavily on manual comparison and keyword-based retrieval, which can miss indirect relationships.
- RAG-based methods improve document question answering by retrieving relevant evidence before generating responses, but standard RAG may not explicitly model policy relationships.
- Graph-based and Graph RAG approaches represent relationships between documents or passages, improving multi-hop retrieval and cross-reference tracing.
- Recent research highlights the need for explainability and reliable evidence grounding in high-stakes enterprise and regulatory applications.

Summary of Literature

- Existing approaches provide strong capabilities for document retrieval, semantic search, and graph-based knowledge representation.
- However, they generally focus on retrieval or question answering rather than systematic propagation of policy-version changes to affected policies.
- A research gap exists in combining change detection, relationship-aware impact propagation, evidence retrieval, confidence scoring, and explainable generation in one framework.

Additional Information

- **Dataset:** ObliQA structured ADGM regulatory documents, augmented with controlled policy-version changes and impact annotations.
- **Core technologies:** React, Python, FastAPI, LangChain, LLM, Sentence Transformers, FAISS, NetworkX, PostgreSQL, and Docker.
- **Project flow:** Change Detection → PolicyGraph → Impact Propagation → RAG Evidence → Impact Ranking → LLM Explanation.
- **Target users:** Target users include compliance officers, policy analysts, risk managers, and governance teams who need traceable and explainable policy-impact insights.

Modules to Cover

- **Policy Document Processing**

- Upload and extract text from policy PDFs
- Split documents into relevant clauses/sections

- **RAG & Semantic Retrieval**

- Create embeddings for policy content
- Retrieve relevant policies and clauses for a given change

- **Policy Change Analysis**

- Compare the changed requirement with existing policies
- Identify related and potentially affected clauses

- **Policy Dependency Graph**

- Represent relationships between policies and clauses
- Trace dependencies and propagate the impact through connected policies

Modules to Cover

- **Impact & Conflict Analysis**

- Identify affected policies, dependencies and conflicts
- Determine the extent of the policy change impact

- **LLM-Based Reasoning**

- Analyze retrieved policy context
- Connect the change with its related requirements and dependencies

- **Explainable Report Generation**

- Provide evidence-backed explanations
- Summarize affected policies and required actions

- **User Interface & Data Management**

- Allow users to upload policies and enter changes
- Display impact analysis, dependency relationships and reports
- Store policy documents and analysis results